

Further Finding the Equation of a Line

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Extend the tables below so that $x = 0$ is shown. Then solve for the line equation.

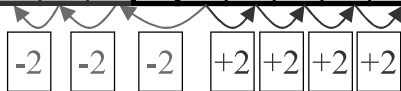


Example

x	3	4	5	6	7
y	9	11	13	15	17

Rewrite taking x back to zero.

0	1	2	x	3	4	5	6	7
3	5	7	y	9	11	13	15	17



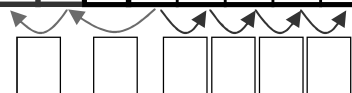
At $x = 0$ Gradient: $+2$

$y = +3$

Equation: $y = 2x + 3$

1

		x	2	3	4	5	6
		y	2	5	8	11	14



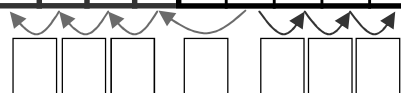
At $x = 0$ Gradient:

$y =$

Equation:

2

			x	4	5	6	7
			y	11	13	15	17



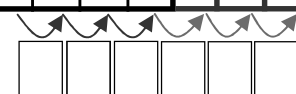
At $x = 0$ Gradient:

$y =$

Equation:

3

x	-6	-5	-4	-3			
y	-9	-6	-3	0			



At $x = 0$ Gradient:

$y =$

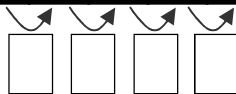
Equation:



These tables have x -values that increase in larger steps. So divide to find the gradient.

4

x	-4	-2	0	2	4
y	-19	-11	-3	5	13



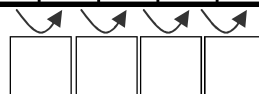
At $x = 0$ Gradient: $\div$

$y =$ Gradient:

Equation:

5

x	0	5	10	15	20
y	-3	22	47	72	97



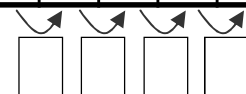
At $x = 0$ Gradient: $\div$

$y =$ Gradient:

Equation:

6

x	-6	-3	0	3	6
y	8	-1	-10	-19	-28



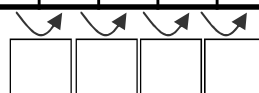
At $x = 0$ Gradient: $\div$

$y =$ Gradient:

Equation:

7

x	0	2	4	6	8
y	-2	-14	-26	-38	-50



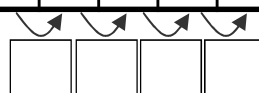
At $x = 0$ Gradient: $\div$

$y =$ Gradient:

Equation:

8

x	-40	-30	-20	-10	0
y	195	155	115	75	35



At $x = 0$ Gradient: $\div$

$y =$ Gradient:

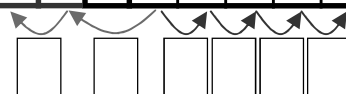
Equation:



These are harder, they require the table to be extended and the x -value steps to be considered.

9

		x	4	6	8	10	12
		y	17	23	29	35	41



At $x = 0$ Gradient: $\div$

$y =$ Gradient:

Equation:

10

			x	12	15	18	21
			y	16	25	34	43

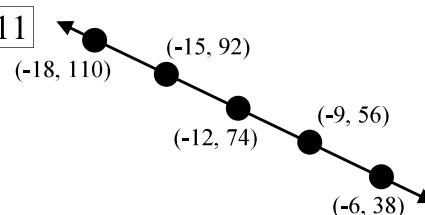


At $x = 0$ Gradient: $\div$

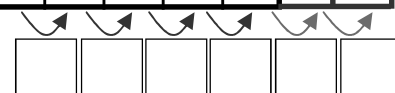
$y =$ Gradient:

Equation:

11



x							
y							



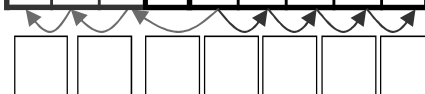
At $x = 0$ Gradient: $\div$

$y =$ Gradient:

Equation:

12

		x	15	20	25	30	35
		y	20	30	40	50	60



At $x = 0$ Gradient: $\div$

$y =$ Gradient:

Equation:

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Extend the tables below so that $x = 0$ is shown. Then solve for the line equation.



Example

x	3	4	5	6	7
y	9	11	13	15	17

Rewrite taking x back to zero.

0	1	2	x	3	4	5	6	7
3	5	7	y	9	11	13	15	17

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At $x = 0$ Gradient: $+2$
 $y = +3$

Equation: $y = 2x + 3$

1

0	1	x	2	3	4	5	6
-4	-1	y	2	5	8	11	14

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 $\searrow$ $\searrow$ $\searrow$ $\searrow$ $\searrow$ $\searrow$

At $x = 0$ Gradient: $+3$
 $y = -4$

Equation: $y = 3x - 4$

2

0	1	2	3	x	4	5	6	7
3	5	7	9	y	11	13	15	17

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 $\searrow$ $\searrow$ $\searrow$ $\searrow$ $\searrow$ $\searrow$ $\searrow$

At $x = 0$ Gradient: $+2$
 $y = 3$

Equation: $y = 2x + 3$

3

x	-6	-5	-4	-3	-2	-1	0
y	-9	-6	-3	0	3	6	9

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 $\searrow$ $\searrow$ $\searrow$ $\searrow$ $\searrow$ $\searrow$

At $x = 0$ Gradient: $+3$
 $y = 9$

Equation: $y = 3x + 9$



These tables have x -values that increase in larger steps. So divide to find the gradient.

4

x	-4	-2	0	2	4
y	-19	-11	-3	5	13

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At $x = 0$ Gradient: $8 \div 2$
 $y = -3$ Gradient: 4

Equation: $y = 4x - 3$

5

x	0	5	10	15	20
y	-3	22	47	72	97

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At $x = 0$ Gradient: $5 \div 5$
 $y = -3$ Gradient: 1

Equation: $y = x - 3$

6

x	-6	-3	0	3	6
y	8	-1	-10	-19	-28

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At $x = 0$ Gradient: $-9 \div 3$
 $y = -10$ Gradient: -3

Equation: $y = -3x - 10$

7

x	0	2	4	6	8
y	-2	-14	-26	-38	-50

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 $\searrow$ $\searrow$ $\searrow$ $\searrow$

At $x = 0$ Gradient: $-12 \div 2$
 $y = -2$ Gradient: -6

Equation: $y = -6x - 2$

8

x	-40	-30	-20	-10	0
y	195	155	115	75	35

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At $x = 0$ Gradient: $-40 \div 10$
 $y = 35$ Gradient: -4

Equation: $y = -4x + 35$



These are harder, they require the table to be extended and the x -value steps to be considered.

9

0	2	x	4	6	8	10	12
5	11	y	17	23	29	35	41

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At $x = 0$ Gradient: $6 \div 2$
 $y = 5$ Gradient: 3

Equation: $y = 3x + 5$

10

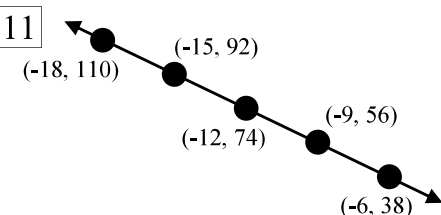
0	3	6	9	x	12	15	18	21
-20	-11	-2	7	y	16	25	34	43

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At $x = 0$ Gradient: $9 \div 3$
 $y = -20$ Gradient: 3

Equation: $y = 3x - 20$

11



x	-18	-15	-12	-9	-6	-3	0
y	110	92	74	56	38	20	2

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At $x = 0$ Gradient: $-18 \div 3$
 $y = 2$ Gradient: -6

Equation: $y = -6x + 2$

12

0	5	10	x	15	20	25	30	35
-10	0	10	y	20	30	40	50	60

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At $x = 0$ Gradient: $10 \div 5$
 $y = -10$ Gradient: 2

Equation: $y = 2x - 10$